



Solve with us 3.1 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

Definition of a vector space V : Definition of a vector space V :

- A vector space V over R is a set along with two functions:

1. $+: V \times V \rightarrow V$: $V \times V \rightarrow V$ (V is closed under vector addition i.e., if $x, y \in V \Rightarrow x + y \in V, \forall x, y \in V, x, y \in V \Rightarrow x + y \in V, \forall x, y \in V$)
 2. $.: R \times V \rightarrow V$: $R \times V \rightarrow V$ (V is closed under scalar multiplication i.e., if $x \in V$ and $c \in R \Rightarrow c.x \in V, \forall x \in V$, and $\forall c \in R$
 $x \in V$ and $c \in R \Rightarrow c.x \in V, \forall x \in V$, and $\forall c \in R$)
- Axioms of vector additions of a vector space
1. $v_1 + v_2 = v_2 + v_1, \forall v_1, v_2 \in V, v_1 + v_2 = v_2 + v_1, \forall v_1, v_2 \in V$.
 2. $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3, \forall v_1, v_2$ and $v_3 \in V, v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3, \forall v_1, v_2$ and $v_3 \in V$.
 3. There exists an element 0 (called the zero vector of V) in V such that $0 + v = v, \forall v \in V$
 $0 + v = v, \forall v \in V$.
 4. There exists a vector v' in V such that $v' + v = v + v' = 0, \forall v \in V$
 $+ v = v + v' = 0, \forall v \in V$ (v' is called the inverse (or negative) of v with respect to addition).
- Axioms relating vector addition and scalar multiplication of a vector space (since this is usual multiplication in real numbers, we suppress the "." sign).
1. For each vector $v \in V, 1v = v.1v = v$.
 2. For each vector $v \in V$ and for each pair $a, b \in R, (a + b)v = av + bv. a, b \in R,$
 $(a + b)v = av + bv$.
 3. For each $a \in R$ and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2. v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$.
 4. For each vector $v \in V$ and for each pair $a, b \in R, (ab)v = a(bv). a, b \in R, (ab)v = a(bv)$.

Consider a set $V = \{(1, -y) \mid y \in R\} \subseteq R^2, V = \{(1, -y) \mid y \in R\} \subseteq R^2$ with the usual addition and scalar multiplication as in R^2 . Answer the following questions (1) and (2).

1 point

Which of the following option(s) is (are) true?

[Hint:Hint: Check the axioms of vector space.]

☐

$(1, 2) \in V, (2, 2) \in V$.

☐

$(2, 3) \in V, (2, 3) \in V$.

☐

$V = R^2, V = R^2$.

☐

$v_1 = (1, -5)$ and $v_2 = (1, 7) \in V \Rightarrow v_1 + v_2 \in V, v_1 = (1, -5)$ and $v_2 = (1, 7) \in V \Rightarrow v_1 + v_2 \in V$.

☐

$v = (1, 9) \in V, c \in R \Rightarrow cv \in V, \forall c \in R, v = (1, 9) \in V, c \in R \Rightarrow cv \in V, \forall c \in R$.





For all $v \in V, cv \in V$ For all $v \in V, cv \in V$ only for $c = 1c = 1$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2) \in V(1, 2) \in V$.

For all $v \in V, cv \in V$ For all $v \in V, cv \in V$ only for $c = 1c = 1$.

1 point

Which of the following option(s) is (are) true?



Zero element does not exist in VV .



Zero element of VV is $(0,0)$.

☐ Inverse element of $(1, 5)$ is $(-1, -5)$.

☐ Inverse element of $(1, 5)$ is $(1, -5)$.

☐ Inverse element of $(1, 5)$ does not exist.



Inverse element of any element of VV does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Zero element does not exist in VV .

Inverse element of $(1, 5)$ does not exist.

Inverse element of any element of VV does not exist.

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ with the

usual addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$, $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$,

where $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ where $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ and scalar multiplication: $c(x, y, z) = (cx, cy, cz)$, $c \in R$ and $(x, y, z) \in V$. $c(x, y, z) = (cx, cy, cz)$, $c \in R$ and $(x, y, z) \in V$.

Answer the following questions (3) and (4).

1 point

Which of the following option(s) is (are) true?

[Hint:Hint: For a vector $v = (x, y, z)v = (x, y, z)$, check whether the condition $x + y + z = 0$ $x + y + z = 0$ is satisfied or not.]



$(1, 2, -3) \in V \in V$.



$(-1, 2, 1) \in V \in V$.



$V = R^3 V = R^3$.



$(5, -7, 2) \in V(5, -7, 2) \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2, -3) \in V \in V$.

$(5, -7, 2) \in V(5, -7, 2) \in V$.

1 point

Inverse element of $(-2, -4, 6)$ with respect to vector addition is

- ☐ $(2, 4, -6)$.
☐ $(2, -4, -6)$.
☐ $(2, 4, 6)$.
☐ $(-2, 4, -6)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(2, 4, -6)$.

1 point

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ with the usual addition as in R^3 and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (cx, cy, cz) & c \neq 0 \end{cases} \quad (x, y, z) \in V, c \in R$$

$$c = 0 \implies c(x, y, z) = (0, 0, 0) \quad (x, y, z) \in V, c \in R$$

Which of the following option(s) is (are) true?

[Hint: By adding any two elements in V , or by multiplying a scalar with any element in V , check whether the resultant element is in V or not.]

☐

V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$

☐

V is closed under scalar multiplication i.e., $(c \in R, v \in V \implies cv \in V)$

$(c \in R, v \in V \implies cv \in V)$.

☐

V has zero element with respect to addition.

☐

$1.v = v$, where $1 \in R$ and $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$

V has zero element with respect to addition.

1 point

Consider usual addition and scalar multiplication on R^2 and the subset

$V = \{(1, -y) \mid y \in R\} \subseteq R^2$

Which of the following option(s) is (are) true?

[Hint: Take two arbitrary elements $(1, -y_1)$ and $(1, -y_2)$ from V and check the conditions given in the options.]

☐

For each element $v \in V$ and for each pair $a, b \in R$, $(a + b)v = av + bv$

$(a + b)v = av + bv$.

☐

For each $a \in R$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$

$$) = av_1 + av_2.$$



For each element $v \in V$ and for each pair $a, b \in R$, $(ab)v = a(bv)$, $a, b \in R$, $(ab)v = a(bv)$.



The set V with the usual addition and scalar multiplication in R^2 is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each element $v \in V$ and for each pair $a, b \in R$, $(a + b)v = av + bv$, $a, b \in R$,

$$(a + b)v = av + bv.$$

For each $a \in R$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$, $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.

For each element $v \in V$ and for each pair $a, b \in R$, $(ab)v = a(bv)$, $a, b \in R$, $(ab)v = a(bv)$.

1 point

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ with the usual addition as in R^3 and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (cx, cy, cz) & c \neq 0 \end{cases} \quad (x, y, z) \in V, c \in R$$

$$c = 0 \implies (0, 0, 0) \in V, c \in R$$

Which of the following option(s) is (are) true?



For each element $v \in V$ and for each pair $a, b \in R$, $(a + b)v = av + bv$, $a, b \in R$,

$$(a + b)v = av + bv$$



For each $a \in R$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$, $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.



For each element $v \in V$ and for each pair $a, b \in R$, $(ab)v = a(bv)$, $a, b \in R$, $(ab)v = a(bv)$.



V is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each $a \in R$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$, $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.

Key Points: 2

- Definition of a subspace: A subset W of a vector space V is a subspace of V if it is a vector space with respect to the same operations defined on V .
- Criteria to check a subset W of a vector space V is a subspace:
 - Zero vector should be in W i.e., $0 \in W$.
 - If two vectors v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
 - Let $c \in R$ and if $v \in W$, then $cv \in W$.

1 point

Let $W = \{(x, y) \mid x + 3y = 1, x, y \in \mathbb{R}\} \subseteq \mathbb{R}^2$. Which of the following options are true?

☐

Zero element exists in W .

☐

If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.

☐

Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$.

☐

W is not a subspace of $\mathbb{R}^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.

W is not a subspace of $\mathbb{R}^2$.

1 point

Let $W = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a diagonal matrix}\}$.

Which of the following option(s) is (are) true?

☐

Zero vector does not exist in W .

☐

If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.

☐

Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$.

☐

W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.

Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$.

W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Key Points: 3

- Linear combination: Let V be a vector space and $v_1, v_2, \dots, v_k \in V$. The linear combination of $v_1, v_2, \dots, v_k$ with coefficients $a_1, a_2, \dots, a_k \in \mathbb{R}$ is the vector $v = \sum_{i=1}^k a_i v_i$.
- A vector $v \in V$ is said to be a linear combination of $v_1, v_2, \dots, v_k \in V$ if there exist $a_1, a_2, \dots, a_k \in \mathbb{R}$ so that $v = \sum_{i=1}^k a_i v_i$.

1 point

Which of the following vectors are linear combinations of $(2, 1)$ and $(6, 3)$ with usual addition and scalar multiplication in $\mathbb{R}^2$?

☐ $(2, 1) + 5(6, 3)$
☐ $(2, 1)$
☐ $(6, 3)$
☐ $(3, 6)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(2, 1) + 5(6, 3)$

$(2, 1)$

$(6, 3)$

1 point

Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ i.e., $v_1 = av_2 + bv_3 + cv_4$, then which of the following options are true?

☐

If $a \neq 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.

☐

If $a = 0$ and $b \neq 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.

☐

v_2 is a linear combination of v_2 .

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $a \neq 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.

If $a = 0$ and $b \neq 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.

v_2 is a linear combination of v_2 .